

Chapter 9

Organic Chemistry: Hydrocarbon and Isomerism

Learning Objectives

- Alkane and cycloalkane
- Alkene
- Skeletal structure
- IUPAC Nomenclature
- Double bond equivalent
- Isomerism
- Z/E configuration

Organic Chemistry and Organic Compounds

- Organic Chemistry is best defined as the chemistry of carbon containing compounds.
- In addition, organic molecules always contain **hydrogen** and often contain atoms of **oxygen, nitrogen, phosphorus, sulfur or halogens**.
- The simplest class of organic compound is **hydrocarbons** which consist of carbon and hydrogen only.
- There are four major classes of hydrocarbons. They are **alkanes, alkenes, alkynes and aromatic hydrocarbons (arenes)**.

Part 1

Alkane and Cycloalkane

Alkane

- Alkanes are saturated organic compounds with the general molecular formula C_nH_{2n+2} .
- The simplest alkane is methane.
- The names of alkanes end with ‘-ane’
- Alkane in which the carbons form a continuous chain with no branches is called **straight chain alkane**.
- The names and properties of some straight chain alkanes are given in the following table.

TABLE 2.7 The first 10 straight-chain alkanes

Memorize
the names

C11: undecane

C12: dodecane

- Substituents are named by changing '–ane' to '–yl'
- meth**ane** → methyl
- but**ane** → butyl

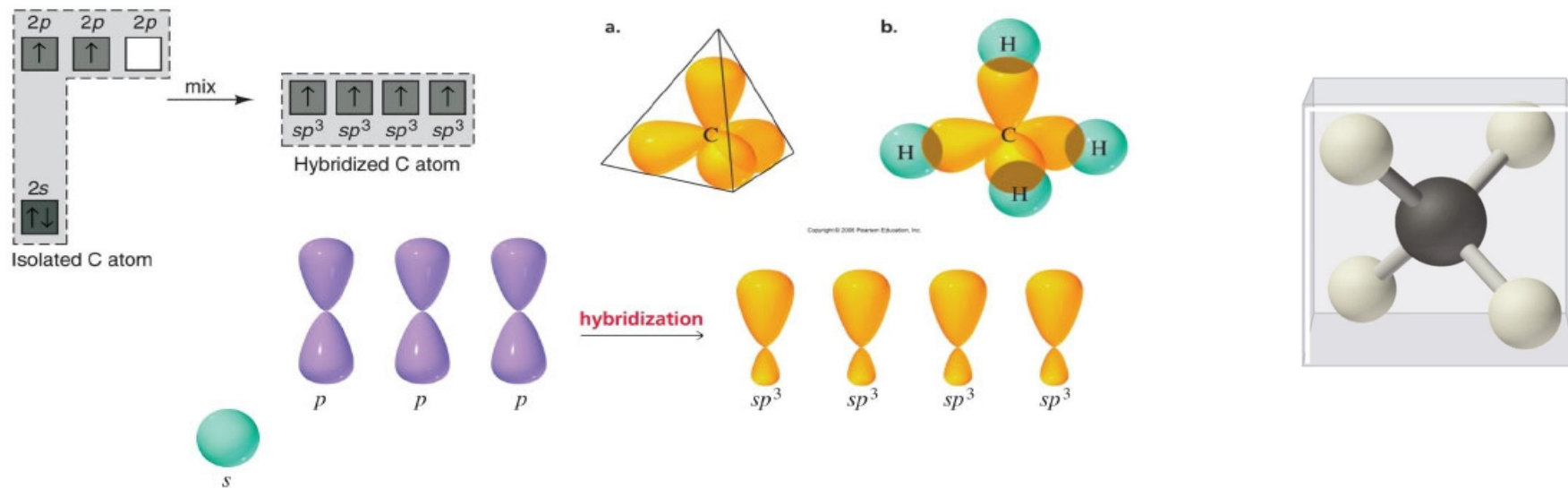
abbreviation

methyl	–CH ₃	Me
ethyl	–CH ₂ CH ₃	Et
propyl	–CH ₂ CH ₂ CH ₃	Pr
butyl	–(CH ₂) ₃ CH ₃	Bu

Name	Number of Carbon Atoms	Molecular Formula	Condensed Structural Formula	Boiling Point (°C)	Uses
methane	1	CH ₄	CH ₄	–162	natural gas constituent
ethane	2	C ₂ H ₆	CH ₃ CH ₃	–89	natural gas constituent
propane	3	C ₃ H ₈	CH ₃ CH ₂ CH ₃	–42	bottled gas
butane	4	C ₄ H ₁₀	CH ₃ CH ₂ CH ₂ CH ₃ or CH ₃ (CH ₂) ₂ CH ₃	0	lighters, bottled gas
pentane	5	C ₅ H ₁₂	CH ₃ (CH ₂) ₃ CH ₃	36	solvent, gasoline
hexane	6	C ₆ H ₁₄	CH ₃ (CH ₂) ₄ CH ₃	69	solvent, gasoline
heptane	7	C ₇ H ₁₆	CH ₃ (CH ₂) ₅ CH ₃	98	solvent, gasoline
octane	8	C ₈ H ₁₈	CH ₃ (CH ₂) ₆ CH ₃	126	gasoline
nonane	9	C ₉ H ₂₀	CH ₃ (CH ₂) ₇ CH ₃	151	gasoline
decane	10	C ₁₀ H ₂₂	CH ₃ (CH ₂) ₈ CH ₃	174	kerosene

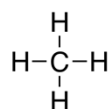
Bonding in Alkane

- Electronic configuration of hydrogen is $1s^1$.
- Electronic configuration of carbon is $1s^2 2s^2 2p^2$.
- Orbital hybridization for carbon gives four sp^3 hybrid orbitals.

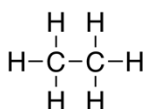


Full Structure vs Skeletal structure

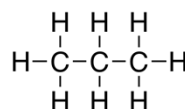
- As we can see, it quickly becomes tedious to draw out full structures, especially for large organic molecules.
- A short-hand method is needed.



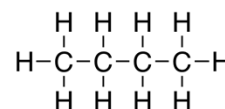
methane



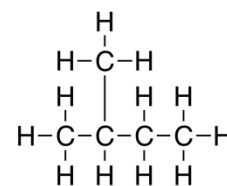
ethane



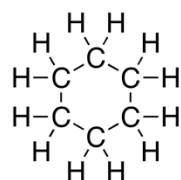
propane



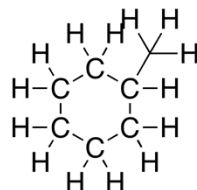
butane



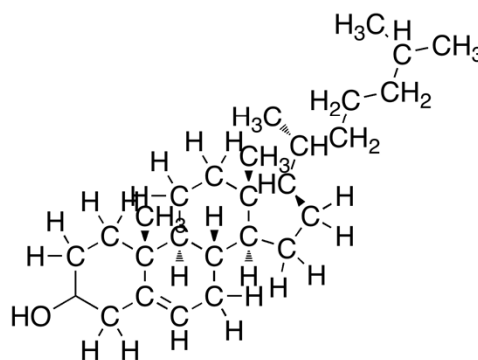
2-methylbutane



cyclohexane

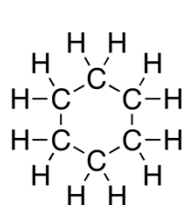


methylcyclohexane

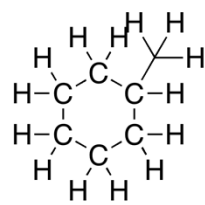


cholesterol

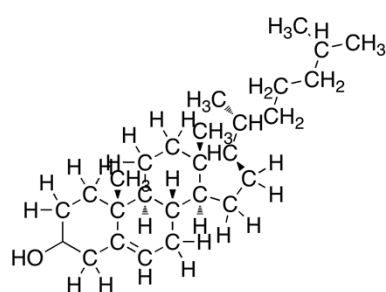
Graphical Representation of Skeletal Structures



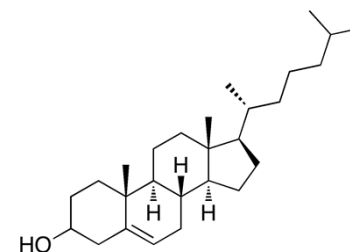
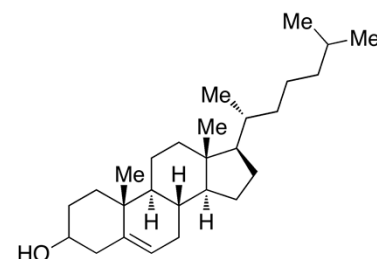
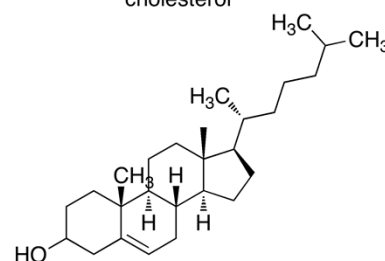
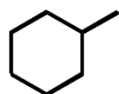
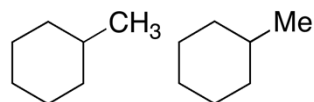
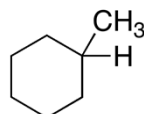
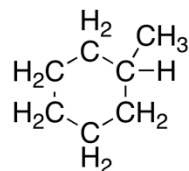
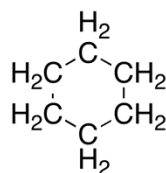
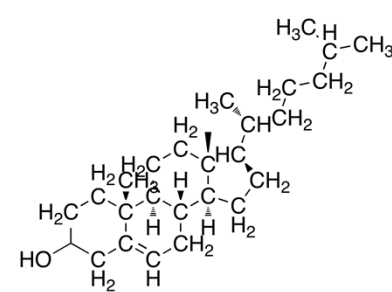
cyclohexane



methylcyclohexane



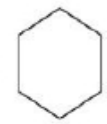
cholesterol



Cycloalkane

- If the ends of a hydrocarbon chain are connected to form a ring of atoms, then it becomes a cyclic hydrocarbon, e.g. cycloalkane.
- Cycloalkanes are named by adding the prefix 'cyclo-' to the name of the alkane. The general molecular formula is C_nH_{2n} (different from C_nH_{2n+2}).

TABLE 2.8 The simple cyclic alkanes

Name	Molecular Formula	Structural Formula	Name	Molecular Formula	Structural Formula
cyclopropane	C_3H_6	$\begin{array}{c} \text{CH}_2 \\ \diagup \quad \diagdown \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \diagdown \quad \diagup \\ \text{CH}_2 \end{array}$ or 	cyclopentane	C_5H_{10}	$\begin{array}{c} \text{H}_2 \\ \\ \text{C} \\ \diagup \quad \diagdown \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \diagdown \quad \diagup \\ \text{H}_2\text{C} \quad \text{CH}_2 \end{array}$ or 
cyclobutane	C_4H_8	$\begin{array}{c} \text{H}_2\text{C} - \text{CH}_2 \\ \quad \\ \text{H}_2\text{C} - \text{CH}_2 \end{array}$ or 	cyclohexane	C_6H_{12}	$\begin{array}{c} \text{H}_2 \\ \\ \text{C} \\ \diagup \quad \diagdown \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \diagdown \quad \diagup \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \\ \text{H}_2 \end{array}$ or 

Part 2

Alkane Nomenclature

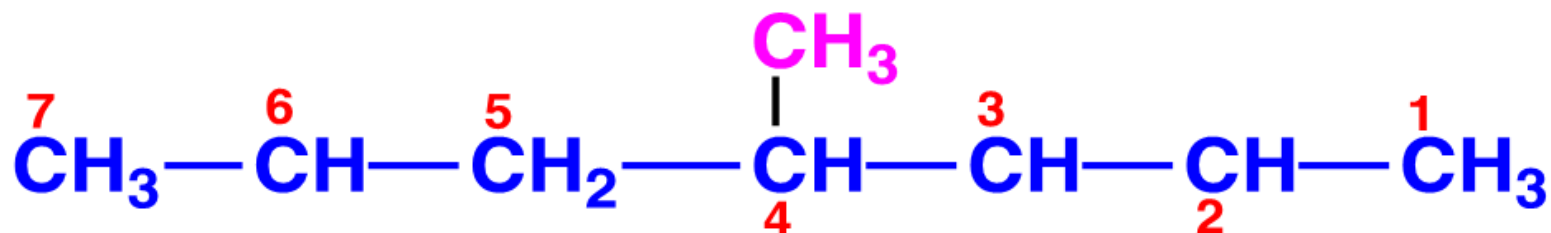
Nomenclature of Alkane

- IUPAC nomenclature helps the scientists to communicate with each other without ambiguity.
- Alkane does not have any functional group, so only chain length is considered.
- IUPAC name of alkanes must fulfil the following requirements:
 1. Maximum length of parent chain
 2. Maximum number of substituents cited as prefix
 3. Lowest locant(s) for the substituent(s) cited as prefix

Naming Alkanes

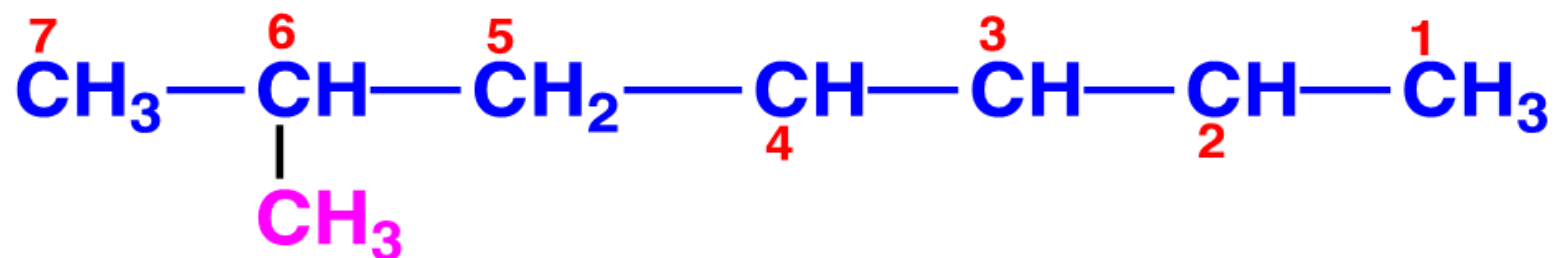
1. Find the longest carbon chain → this is the parent chain (everything outside this parent chain are substituents)
2. Write the substituents (methyl, ethyl, ...) in front of the parent chain, in alphabetical order, e.g. **b**utyl, **e**thyl, **m**ethyl, **p**ropyl
3. If there are more than one identical substituents, use the following prefix for the number: di(2), tri(3), tetra(4), penta(5), hexa(6), ...
4. The parent chain is numbered as C1, C2, C3, ... and the locant number of substituents are cited before the substituents, separated by a hyphen (-)
5. The locant numbers must be kept as small as possible

Examples (1)



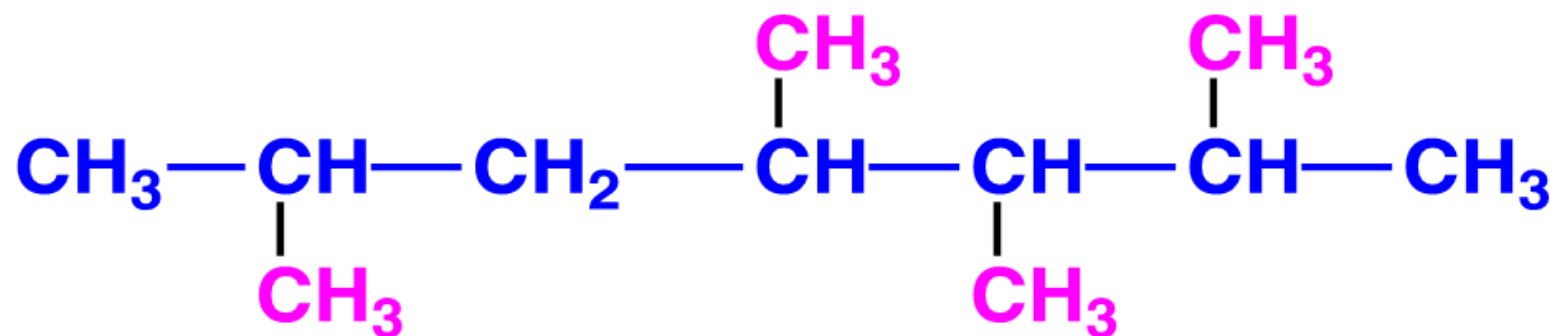
- Find longest chain → this is the parent chain
 - C₇ = heptane → parent name
- Find substituent(s)
 - Only one carbon is outside the parent chain
 - C₁ = methane → substituent = methyl → prefix
- Name is methylheptane [no space between alphabets]
- Lastly, add locant number of the prefix
 - 4-methylheptane (*from either direction, methyl is on C4*)

Examples (2)



- Find the longest chain
- Find the substituent(s)
- Give name

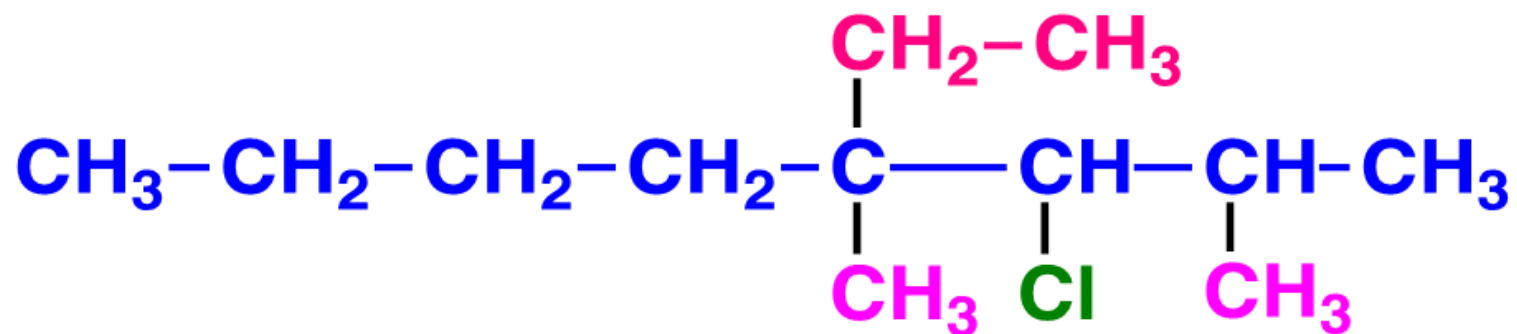
Examples (3)



- Find the longest chain
- Find the substituent(s) – how to deal with multiple substituents?
- Give name

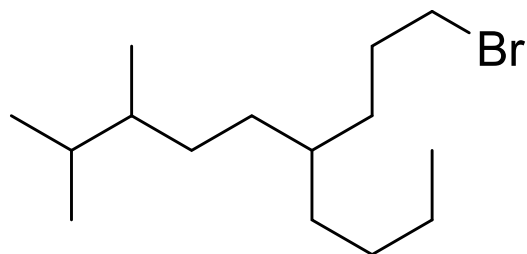
Example (4)

- Haloalkanes follow the same nomenclature logic: halo groups (fluoro, chloro, bromo and iodo) are always treated as substituents (prefix).



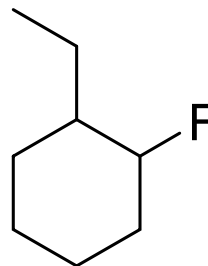
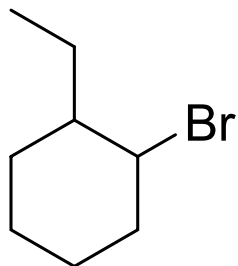
Example (5)

- Substituent in substituent



Example (6)

- Cycloalkane
- Additional rule (apply to alkane, cycloalkane and any other classes of organic compounds): lowest locant number for the first cited substituent



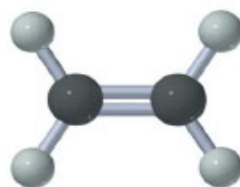
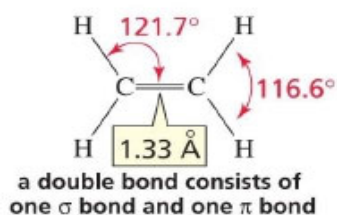
Part 3

Alkene

Alkene

- The class of organic compounds that contains at least one C=C bond are called alkenes, with general formula C_nH_{2n} .
- The names of the alkenes end with '-ene'.
- The simplest alkene is ethene, $H_2C=CH_2$

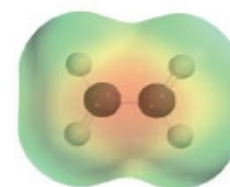
What is the hybridization of the carbon?
What is the geometry?



ball-and-stick model of ethene



space-filling model of ethene



electrostatic potential map for ethene

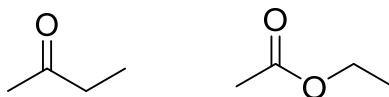
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Double Bond Equivalent (DBE)

- The general formula of an alkane is C_nH_{2n+2} .
 - Why?
- If one C-C bond of the alkane is changed to a C=C, two hydrogen atoms is removed from the molecule $\rightarrow$ formula becomes C_nH_{2n} .
- If two ends of the carbon chain is joined by a C-C bond (hence forming a ring), two hydrogen atoms is also removed $\rightarrow$ formula also becomes C_nH_{2n} .
- C=C and ring is equivalent since each of them results in a reduction of two H in the formula. Similarly, one $C\equiv C$ bond is equivalent to two C=C bonds.
- Double Bond Equivalent: sum of π -bonds and rings in a molecule (C=C and C=O contributes 1 to DBE; $C\equiv C$ and $C\equiv N$ contributes 2 to DBE).

Application of DBE

- $DBE = \frac{C \times 2 + 2 - H - X + N}{2}$
- C, H, **X** and N refer to the number of carbon, hydrogen, **halogen** and nitrogen in the molecular formula.
- Oxygen and sulfur are absent in the equation, since their number has no impact on DBE.



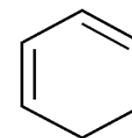
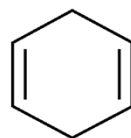
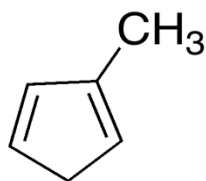
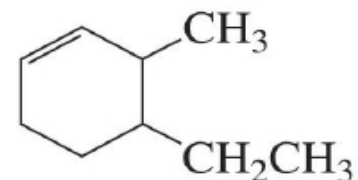
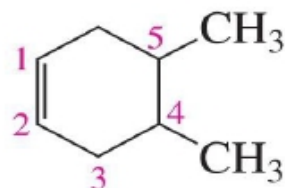
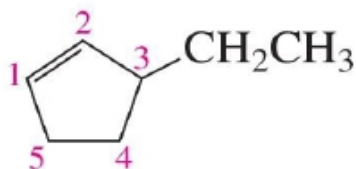
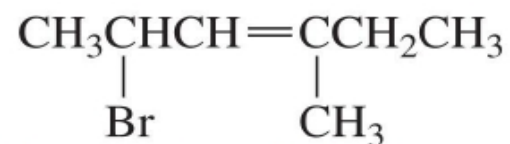
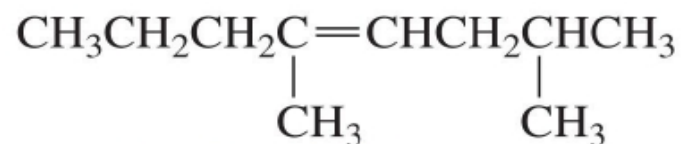
- Molecular formula $\rightarrow$ DBE $\rightarrow$ information about π -bonds and rings
- Given a structure $\rightarrow$ count number of C/O/N/X $\rightarrow$ calculate number of H

Nomenclature of Alkene

- IUPAC rule for alkene nomenclature:
 1. Maximum number of C=C bond(s) in the parent chain
 2. Maximum length of parent chain
 3. Lowest locant(s) for C=C bond(s)
 4. Maximum number of substituents cited as prefix
 5. Lowest locant(s) for the substituent(s) cited as prefix
 6. Lowest locant for the first cited substituent

Compounds with two, three, ... C=C bonds
are named as “diene”, “triene”, ...

Examples of Alkene Nomenclature

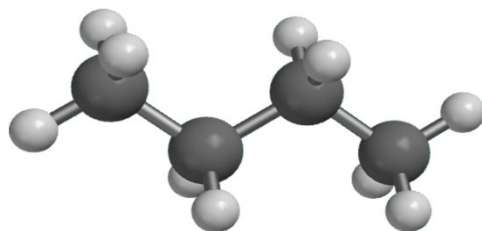


Part 4

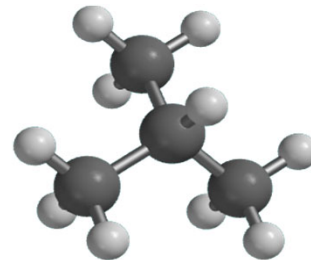
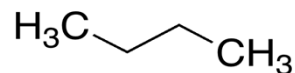
Isomerism

Isomer

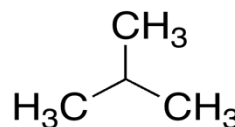
- Isomers are compounds that have the same molecular formula but different arrangement of atoms.
- Isomers may have different carbon skeletons or different functional groups or different locations of a functional group along the chain.



n-butane, C₄H₁₀



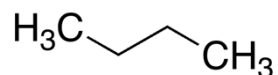
isobutane, C₄H₁₀
(2-methylpropane)



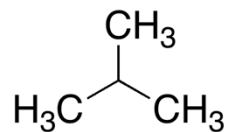
Alkane Isomer

- Isomers usually have different properties:

n-butane, C₄H₁₀

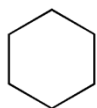


isobutane, C₄H₁₀
(2-methylpropane)

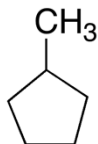


Physical Properties	<i>n</i> -butane	<i>iso</i> -butane
Boiling point	0 °C	−12 °C
Melting point	−138 °C	−159 °C
Density of liquid	0.622 g/mL	0.604 g/mL

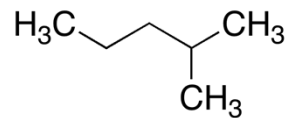
- Other examples:



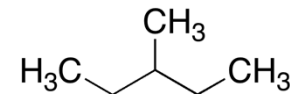
cyclohexane
C₆H₁₂



methylcyclopentane
C₆H₁₂



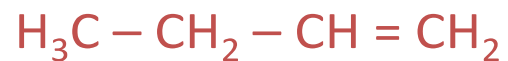
2-methylpentane
C₆H₁₄



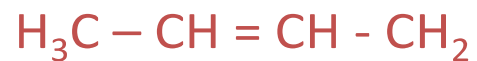
3-methylpentane
C₆H₁₄

Alkene Isomer (1)

- Positional Isomer



1-butene

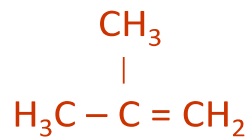


2-butene

- Skeleton Isomer



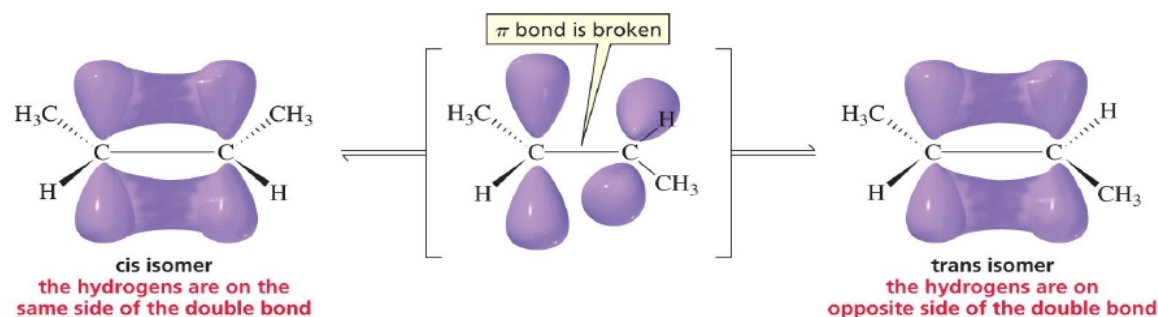
1-butene



Methylpropene

Alkene Isomer (2)

- Cis-trans isomer
 - Rotation about a double bond does not readily occur.
 - If rotation were to occur, the two p orbitals would cease to overlap and the pi bond would break.

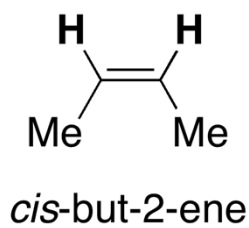


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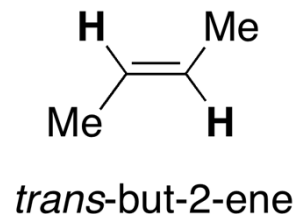
- Cis-trans isomer is a sub-category of stereoisomer.
 - Another subcategory of stereoisomer is enantiomer.

Alkene Isomer (3)

- Due to the rigidity of the C=C bond (unable to freely rotate), the alkenes may have two stereoisomers – cis and trans



Two H on the
same side → *cis*
You can also say
two Me on the
same side → *cis*



cis and trans isomers are not possible for these compounds because two substituents on an sp^2 carbon are the same

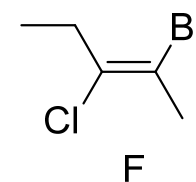
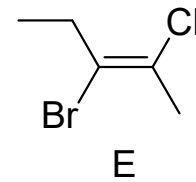
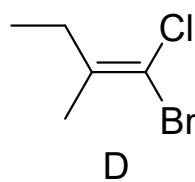
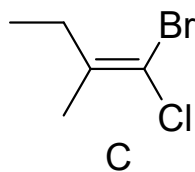
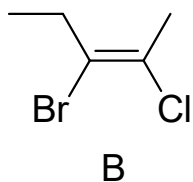
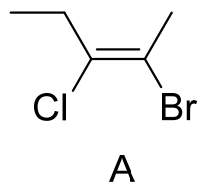


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When we name an alkene as “*cis*” or “*trans*”, we are referring to two **identical groups** on **each** C=C carbon.

Z-E Nomenclature System

- Cis-trans nomenclature has its limitation:
 - Two **identical groups** on **each** C=C carbon
 - What happens if there are no identical groups?
- Which two are one pair of “cis-trans” isomers?
- Can you name them as “cis” or “trans”?

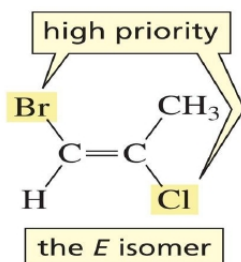
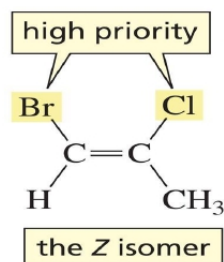


Cahn-Ingold-Prelog (CIP) Rule

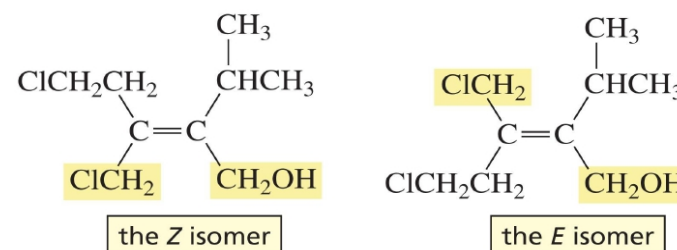


- In order to name an alkene as “Z” or “E”, we need to compare the “priority” of the two groups attached to the same carbon by the CIP rule.
 1. Compare the atomic number of the two atoms which are directly attached to the C=C carbon. For ease of discussion, these are the “level 1” atoms. The atom with higher atomic number has higher priority, e.g. Cl > O > N > C > H.
 2. If level 1 atoms are isotopes of the same element, compare their mass number. The isotope with higher mass number has higher priority, e.g. D > H.
 3. If there is a tie for the “level 1 atoms”, continue the same comparison on “level 2 atoms”, which are the atoms directly bonded to the “level 1 atoms”. Continue on “level 3”, “level 4” ... if needed.
- Once the priority of two groups on each C=C carbon is determined:
 - Z-isomer: two “high priority” groups are on the same side of the C=C bond
 - E-isomer: two “high priority” groups are on the opposite side of the C=C bond

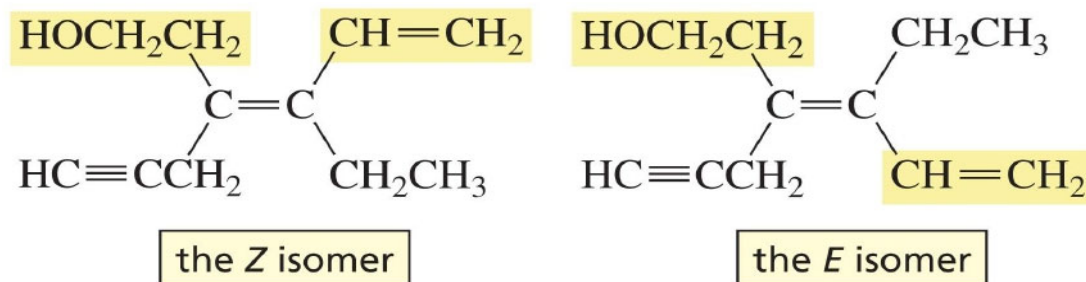
Examples of Z-E Isomer



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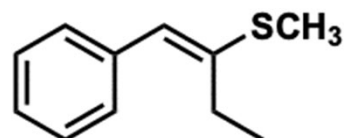
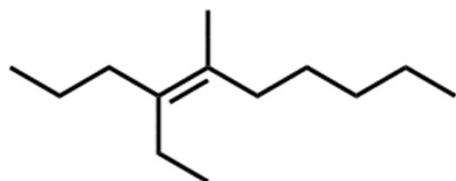
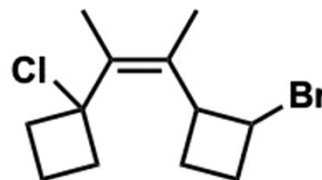
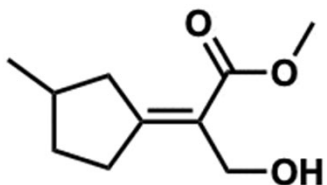


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How about double and triple bonds?

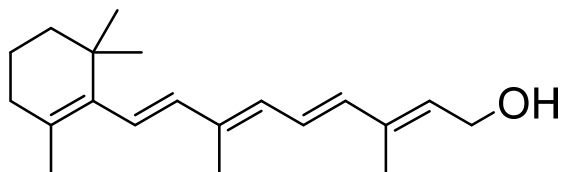
Exercise: Z/E nomenclature

- Determine the Z/E configuration of the following alkenes:



Exercise: DBE and Isomerism

- Calculate the DBE of C_4H_6 and hence draw all isomers.
- Count the number of C/O and DBE of vitamin A. Hence, calculate number of hydrogen and write its molecular formula.



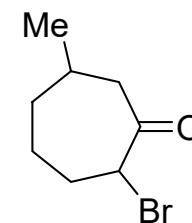
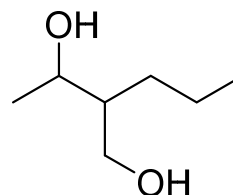
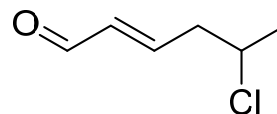
Part 5

Nomenclature of other organic compounds

Class of Compound	General Formula	Functional Group	Suffix	Example
Alcohol	R-OH	-OH (hydroxyl)	-ol	Propanol
Aldehyde	R-CHO	-CO- (carbonyl)	-al	Propanal
Ketone	R ¹ -CO-R ²	-CO- (carbonyl)	-one	Propanone
Carboxylic acid	R-COOH	-COOH (carboxyl)	-oic acid	Propanoic acid
Arene	R-C ₆ H ₅	Benzene	Benzene	Diethylbenzene
Phenol	R-C ₆ H ₄ -OH	Phenol	Phenol	2-Chlorophenol

- The nomenclature of other classes of organic compounds is beyond the bridging course:

- Ether
- Ester
- Amide
- Amine
- Nitrile



A more “complete” IUPAC Nomenclature rule

1. Maximum number of principle functional groups cited as suffix
2. Maximum number of C=C and C≡C bonds together
3. Maximum length of parent chain/ring
4. Maximum number of C=C bonds
5. Lowest locant(s) for principle groups(s)
6. Lowest locant(s) for multiple bonds
7. Lowest locant(s) for double bond(s)
8. Maximum number of substituents cited as prefix
9. Lowest locant(s) for the substituent(s) cited as prefix
10. Lowest locant for the first cited substituent